

Rejection sampling: target $N(0, 1)$

① Cauchy proposal $g(x) = 1/(\pi(1+x^2))$ — first attempt

$$\frac{f}{g} = \frac{\frac{1}{\sqrt{2\pi}}e^{-x^2/2}}{\frac{1}{\pi(1+x^2)}} = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}\pi(1+x^2) = e^{-x^2/2}x^2\pi \quad \text{[E1]}$$

[E1]: I made $\pi(1+x^2)$ into: $x^2\pi$ and removing the constant factor.

$$M = \sup_x e^{-x^2/2}x^2\pi, \quad \log(e^{-x^2/2}) + \log(x^2) = 0 \iff -\frac{x^2}{2} + \log(x^2) = 0 \quad \text{[E2]}, \quad \text{forces } x = 0$$

[E2]: In trying to maximize the function I did not take the derivative just the log and set it equal to 0, lol.

$$\left(-\frac{x^2}{2}\right)' = \frac{1}{2}x^2 = x \quad \text{[E3]}$$

[E3]: Forgetting '-'.

① Redo

$$\begin{aligned} \frac{1}{\sqrt{2\pi}}e^{-x^2/2}\pi(1+x^2) &= \frac{\sqrt{\pi}}{\sqrt{2}}(1+x^2)e^{-x^2/2} \\ \left(\frac{\sqrt{\pi}}{\sqrt{2}}e^{-x^2/2}(1+x^2)\right)' &= 0 \iff \frac{\sqrt{\pi}}{\sqrt{2}}\left((e^{-x^2/2})'(1+x^2) + e^{-x^2/2}(1+x^2)'\right) = 0 \\ &\iff \frac{\sqrt{\pi}}{\sqrt{2}}\left(-xe^{-x^2/2}(1+x^2) + e^{-x^2/2} \cdot 2x\right) = 0 \\ &\iff \frac{\sqrt{\pi}}{\sqrt{2}}\left(e^{-x^2/2}(-x(1+x^2) + 2x)\right) = 0 \\ &\iff \frac{\sqrt{\pi}}{\sqrt{2}}\left(xe^{-x^2/2}(2 - (1+x^2))\right) = 0 \\ &\iff \frac{\sqrt{\pi}}{\sqrt{2}}\left(x(1+x^2)e^{-x^2/2}\right) = 0 \quad \text{[E4]} \end{aligned}$$

[E4]: $2 - (1+x^2) = 1 - x^2$ and NOT $1 + x^2$

Log-odds and posterior

$$\begin{aligned}\log \frac{P(Y = 1 \mid X = x)}{P(Y = 0 \mid X = x)} &= \log P(Y = 1 \mid X = x) - \log P(Y = 0 \mid X = x) \\&= \frac{1}{\sqrt{2\pi}} \log \left(e^{-(x-\mu_1)^2/2} \right) - \frac{1}{\sqrt{2\pi}} \log \left(e^{-(x-\mu_0)^2/2} \right) \quad \text{[E5]} \\&= -\frac{\|x - \mu_1\|^2}{2} + \frac{\|x - \mu_0\|^2}{2} \\&= \frac{(x - \mu_0)^2}{2} - \frac{(x - \mu_1)^2}{2} = \frac{x^2 + \mu_0^2 - 2x\mu_0 - (x^2 + \mu_1^2 - 2x\mu_1)}{2} \\&= \frac{\|\mu_0\|^2 - \|\mu_1\|^2 - 2\langle x, \mu_0 - \mu_1 \rangle}{2} = \frac{0^2 - 8 - 2 \cdot (-2x_1 - 2x_2)}{2} \\&= 2x_1 + 2x_2 - 4\end{aligned}$$

E[5]: $\log \left(\frac{1}{\sqrt{2\pi}} e^{-(x-\mu_1)^2/2} \right) = \log \left(\frac{1}{\sqrt{2\pi}} \right) - \left(\frac{x-\mu_1^2}{2} \right)$ but somehow I decided to drop the factor right after so it ends up magically being correct

Finding the hessian $f(x) = \log(1 + e^{-x})$

While I had certainly heard of the hessian before, I have not for a long time done differentiation like this before:

$$\begin{aligned}(e^{-x})' &= -e^{-x}, \quad \nabla f(x) = \frac{1}{1+e^{-x}} \cdot (-e^{-x}) = -\frac{e^{-x}}{1+e^{-x}} \\ \nabla^2 f(x) &= -1 \cdot \left(\left(\frac{1}{1+e^{-x}} \right)' \cdot e^{-x} + \frac{1}{1+e^{-x}} \cdot (e^{-x})' \right) \\ &= -1 \cdot \left(-\frac{1}{(1+e^{-x})^2} \cdot (-e^{-x}) \cdot e^{-x} + \frac{1}{1+e^{-x}} \cdot (-e^{-x}) \right) \\ &= -1 \cdot \left(\frac{e^{-2x}}{(1+e^{-x})^2} - \frac{e^{-x}}{1+e^{-x}} \right) \\ &= \frac{e^{-x}}{1+e^{-x}} - \frac{e^{-2x}}{(1+e^{-x})^2} = \frac{e^{-x}(1+e^{-x})^2 - (1+e^{-x})e^{-2x}}{(1+e^{-x})^3} \\ &= \frac{e^{-x}(1+2e^{-x}+e^{-2x}) - e^{-2x} - e^{-3x}}{(1+e^{-x})^3} = \frac{e^{-x} + 2e^{-2x} + \cancel{e^{-3x}} - e^{-2x} - \cancel{e^{-3x}}}{(1+e^{-x})^3} \\ &= \frac{e^{-x} + e^{-2x}}{(1+e^{-x})^3} = \frac{e^{-x}(1+e^{-x})}{(1+e^{-x})^3} = \frac{e^{-x}}{(1+e^{-x})^2} \cdot \frac{e^{2x}}{(e^x)^2} \\ &= \frac{e^x}{((1+e^{-x})e^x)^2} = \frac{e^x}{(e^x+1)^2} \geq 0\end{aligned}$$

When I did this I was surprised that it turns out to be correct actually.

Lagrange dual of an LP

Consider this optimization instance:

$$\begin{aligned} \max \quad & c^T x + b^T y \\ \text{s.t.} \quad & Ax + By = d, \quad x \geq 0, \quad y \geq 0, \quad x, y \in \mathbb{R}^n \end{aligned}$$

Defining the langragian for the optimization problem above (never having seen the langrangian defined until a few weeks ago):

$$L(x, y, \mu, \lambda) = c^T x + b^T y + \mu^T (Ax + By - d) + \lambda^T x + \lambda^T y$$

$$g(\mu, \lambda) = \sup_{x, y} L(x, y, \mu, \lambda)$$

$$\begin{aligned} L(x, y, \mu, \lambda) &= c^T x + b^T y + \mu^T Ax + \mu^T By - \mu^T d + \lambda^T x + \lambda^T y \\ &= (c + A^T \mu + \lambda)^T x + (b + B^T \mu + \lambda)^T y - \mu^T d \end{aligned}$$

$$g(\mu, \lambda) = \begin{cases} -\mu^T d & \text{if } c + A^T \mu \leq \lambda \text{ and } b + B^T \mu \leq \lambda \\ +\infty & \text{else} \end{cases}$$

but since $\lambda \geq 0$:

$$g(\mu) = \begin{cases} -\mu^T d & \text{if } c + A^T \mu \leq 0 \text{ and } b + B^T \mu \leq 0 \\ +\infty & \text{else} \end{cases}$$

$$\min \quad -\mu^T d \quad \text{s.t.} \quad c \leq -A^T \mu \quad \text{and} \quad b \leq -B^T \mu$$